Reports

Ecology, 94(8), 2013, pp. 1667–1674
© 2013 by the Ecological Society of America

Synergistic effects of algal overgrowth and corallivory on Caribbean reef-building corals

Alexander T. Wolf[1,4] and Maggy M. Nugues[1,2,3]

[1]*Coral Reef Ecology Group (CORE), Leibniz Center for Tropical Marine Ecology (ZMT), Fahrenheitstrasse 6, 28359 Bremen, Germany*
[2]*Laboratoire d'Excellence "CORAIL" and USR 3278 CRIOBE CNRS-EPHE, CBETM de l'Université de Perpignan, 66860 Perpignan Cedex, France*
[3]*Caribbean Research and Management of Biodiversity (CARMABI), Piscaderabaai z/n, P.O. Box 2090, Willemstad, Curaçao*

Abstract. Indirect biotic interactions play a crucial role in structuring ecological communities, but many of these interactions have not been explored. Algal competition and corallivory are two major stressors contributing to the decline of coral reefs. Here, we provide the first evidence of algal-induced corallivory and synergistic effects between the two stressors on corals. When corals (*Montastraea faveolata*) were placed in contact with algae (*Halimeda opuntia*) together with corallivorous fireworms (*Hermodice carunculata*) in aquaria, corals suffered high tissue mortality. This mortality was reduced in the presence of algae only, and no mortality occurred in the presence of fireworms only or when excluding both algae and fireworms. These findings were supported by field observations showing a predominance of fireworms inside algae contacting live corals, and by an in situ experiment demonstrating higher coral mortality in contact with algae left undisturbed than with algae from which all mobile epifauna were periodically removed. Among the main contributing mechanisms, we suggest that algal contact produces decaying coral tissue that attracts the corallivore and enhances its aggregation behavior. Our study demonstrates an indirect effect pathway by which algae can impact corals, which shares similarities with the classic models of apparent competition and habitat facilitation.

Key words: a*pparent competition; Caribbean; coral; corallivory; disease; habitat facilitation; interaction modification; macroalgae; trait-mediated indirect interactions.*

Introduction

Indirect biotic effects occur when one organism or species impacts another through intermediaries such as shared resources or common enemies (Menge 1995). They are ubiquitous across taxa and habitats (Menge 1995, Chaneton and Bonsall 2000) and can propagate through ecological communities by changes in species abundance through direct consumption, and/or by changes in species traits, such as behavior, morphology, and/or life history (Werner and Peacor 2003). Because indirect effects can alter the strength of, or even reverse, a direct interaction between species, knowledge about indirect links is vital to an understanding of ecological communities, especially today when ecosystems are experiencing a variety of novel perturbations (Wootton 2002). However, although interest in indirect effects has escalated over the past decades (Wootton 2002), many of them remain unexplored.

Coral reefs are in global decline, with benthic algae commonly replacing corals, a trend that is driven by algal overgrowth of live and recently dead corals (McManus and Polsenberg 2004). Coral-algal competition is known to be mediated through complex interactions by various components of the reef community, including herbivores, predators, and microbes (Pennings 1997, McManus and Polsenberg 2004, Barott and Rohwer 2012). However, the role of corallivores in coral-algal competition has not been investigated (Mumby 2009). Corallivory is an important, but often underestimated, biotic stressor for coral reefs (Rotjan and Lewis 2008). In the Indo-Pacific, the seastar *Acanthaster planci* kills more coral than any other disturbances (Fabricius et al. 2010). In the Caribbean, corallivores have not been thought of as constituting a

Manuscript received 27 April 2012; revised 3 April 2013; accepted 17 April 2013. Corresponding Editor: J. F. Bruno.
[4] E-mail: alexthomaswolf@gmail.com

significant cause of coral mortality (Mumby 2009). However, a number of invertebrate corallivores have recently been found to be associated with coral diseases or to act as vectors of coral pathogens, such as the amphinomid fireworm *Hermodice carunculata* (Sussman et al. 2003, Miller and Williams 2006) and the corallivorous snail *Coralliophila abbreviata* (Williams and Miller 2005). Corallivorous invertebrates are common algal dwellers, presumably for food, shelter and/or recruitment. These organisms could influence coral-algal interactions through feeding and transmission of disease, creating an indirect effect pathway by which algae could affect corals.

We investigated the role of the corallivorous fireworm *H. carunculata* in interactions between the green calcareous alga *Halimeda opuntia* and Caribbean reef-building corals. *H. opuntia* provides a complex three-dimensional framework known to host a wide variety of invertebrates, including fireworms (Naim 1988). It is also known to cause coral mortality and disease (Nugues et al. 2004). *H. carunculata* is commonly observed feeding on diseased corals (Miller and Williams 2006) and has been shown to act as a vector of the coral-bleaching pathogen *Vibrio shiloi* in the Mediterranean (Sussman et al. 2003). We hypothesized that corallivory by the worm could enhance mortality in corals interacting with the alga. First, we investigated the abundance of the worm in algae overgrowing different coral species and related its abundance with coral health in the field. Second, we tested the effects of whole algal-associated epifauna on coral bleaching, death and disease in interactions between *H. opuntia* and the coral *Montastraea faveolata* in a field experiment. Finally, we examined the separate and combined effects of the algae and the worm on coral bleaching and mortality under laboratory conditions. This last experiment was also performed using another corallivore, the snail *C. abbreviata*.

Materials and Methods

Field survey.—If *H. carunculata* is using *H. opuntia* to feed on corals, it should be more abundant in algae overgrowing living corals compared to algae exclusively covering dead coral skeleton, and its abundance should be negatively associated with coral health. Additionally, since corallivores commonly exhibit prey preferences (Rotjan and Lewis 2008), their abundance in the algae could differ among coral species. To test these hypotheses, clumps of *H. opuntia* overgrowing live tissue from five coral species (*M. faveolata*, *Montastraea cavernosa*, *Siderastrea siderea*, *Colpophyllia natans*, *Diploria strigosa*) and dead coral skeleton only ($n = 11$–14 replicates) were haphazardly collected using Ziploc bags (SC Johnson, Racine, Wisconsin, USA) by swimming along the reef terrace (4–8 m depth) at Carmabi Buoy One (12°07.551′ N, 68°58.529′ W) in Curaçao, in the southern Caribbean. Clumps overgrowing dead coral skeletons only were collected >1 m away from any living corals. We also collected *H. opuntia* overgrowing the coral *M. faveolata* and algae overgrowing dead coral skeletons only at three other sites on the island (Marie Pompoen, 12°05.525′ N 68°54.516′ W; Spanish Water, 12°03.790′ N, 68°51.190′ W; and Playa Kalki, 12°22.530′ N, 69°09.470′ W). After removing each clump of algae overgrowing live coral tissue, coral health underneath the algae was scored as follows: 1, tissue normal; 2, tissue partially bleached; 3, tissue completely bleached; 4, tissue <50% dead; 5, tissue ≥50% dead. Back in the laboratory, all fireworms were separated from the algal segments, identified microscopically following Fauchald (1977), and counted. Sampling was carried out between March and April 2010.

Field experiment.—We tested the effect of *H. opuntia*-associated epifauna on corals at Carmabi Buoy Zero (12°07.417′ N, 68°58.592′ W) in Curaçao. Colonies of *M. faveolata* were haphazardly chosen swimming along the reef terrace. They were used in the experiment if the colonies showed no sign of disease and were not overgrown by algae. The coral *M. faveolata* was chosen for this study as it is a major reef building species, and adult corals are abundant. Also, our field survey had indicated a high abundance of fireworms in *H. opuntia* in contact with this species. The colonies were tagged and randomly allocated to one of the following five treatments ($n = 20$ colonies each): T1, *H. opuntia* added, a clump (~90 g wet mass) of *H. opuntia* with all epifauna removed in the lab was put inside a nylon mesh bag (1-cm mesh) and positioned to overgrow ~90 cm^2 of living coral surface using rubber bands and nails inserted onto nearby dead coral substrate, with no further manipulation; T2, *H. opuntia* added, epifauna removed, as with T1 but the mesh bag was retrieved every five days, all epifauna was removed from the alga in the lab, and the mesh bag with the alga was returned to its initial position on the same day; T3, *H. opuntia* mimic, a plastic aquarium plant of similar structure and size was used instead of the living alga, the epifauna was not removed during the experiment; T4, procedural control, only a nylon mesh bag was placed in contact with the coral tissue; T5, control, the coral colony was left untouched. The *H. opuntia* mimic treatment was used to distinguish between the biological and physical properties of the alga. In addition, 20 nylon mesh bags with similarly sized clumps of *H. opuntia* were placed onto dead coral substrate >1 m away from any living corals (T6) to control for the effect of coral-algal contact on *H. carunculata*.

The experiment was run for 25 days (20 March–14 April 2010). All replicates were checked every five days. *H. opuntia* clumps remained healthy for the duration of the study. At the end of the experiment, the mesh bags were collected using Ziploc bags. The surface areas of recently dead coral, bleached, and healthy tissue underneath the mesh bag were measured to the nearest cm^2 using a flexible transparent plastic grid divided into 100 1 × 1 cm square cells and were expressed as

percentage of total area covered by the bag. The controls (T5) suffered no mortality and had 100% healthy tissue. In addition, signs of white plague, in the form of recently dead tissue characterized by a sharp and regular demarcation between apparently healthy coral tissue and freshly denuded skeleton often extending beyond the algal overgrown area were recorded. Back in the laboratory, all epifauna were removed from the living and mimic *H. opuntia* and mesh bags. *H. carunculata* were identified and counted.

Laboratory experiment.—The field experiment had its limitations since we could not remove *H. carunculata* without removing the reminder of the epifauna, access by *H. carunculata* could not be controlled and the algae had to be periodically disturbed to remove the epifauna. We therefore performed a laboratory experiment in which we could control the presence/absence of *H. carunculata* without disturbing the algae and study the independent and combined effects of the fireworm and *H. opuntia* on *M. faveolata*. Four outdoor 1 m^3 ($1 \times 1 \times 1$ m) aquaria were each split in two equal sections, provided with flow-through seawater (~4–5 L/minute) subject to natural daily temperature fluctuations (range 29.4° to 30.8°C), and shaded to ensure light conditions similar to the depth range at which all organisms were collected (~30% of the incident surface photosynthetically active radiation). One section of each aquarium was randomly allocated to one of two corallivore treatments: (1) corallivores present, 30 fireworms (~3–6 cm in length, collected in *H. opuntia* clumps) were added, and (2) corallivores absent, no fireworms added. Within each section, six coral fragments of similar size (~300 cm^2) and free of epifauna were placed and randomly allocated to one of two algae treatments: (1) *H. opuntia* present, a clump (~90 g wet mass) of *H. opuntia* with all epifauna removed was placed inside a nylon mesh bag (1 cm mesh) and positioned with cable ties to overgrow ~90 cm^2 of living coral surface, and (2) *H. opuntia* absent, no *H. opuntia* and mesh bag were added (Appendix A: Fig. A1). Coral fragments were 20–25 cm apart. All organisms were collected on the reef terrace at Carmabi Buoy Zero. Coral fragments were removed from the reef by breaking off columns below the live coral to avoid tissue damage and acclimated for five days in the experimental aquaria prior to experimentation. The same experiment was also carried out using the corallivorous snail *C. abbreviata* and the coral *Montastraea annularis*, which is commonly infested by the snail (Baums et al. 2003). In this second experiment, 60 snails were added to each section assigned to the corallivores present treatment.

The experiments were run for 24 days (16 October to 9 November 2010 for *H. carunculata* and 18 November to 12 December 2010 for *C. abbreviata*). Tank surfaces were cleaned every two days with scouring pads. All replicates were checked during the tank cleaning process. *H. opuntia* and corallivores remained healthy for the duration of the study. At the end of the experiment, the mesh bags were collected using Ziploc bags and the surface areas of recently dead coral, bleached, and healthy tissue were measured. Complete fragment mortality was also recorded. Upon complete death, fragments showed rapid tissue disintegration and white plague signs typical of the natural conditions were not observed.

Statistical analyses.—Data from our field survey and field experiment violated parametric assumptions, so nonparametric or semiparametric statistics were used. In the field survey, the biomass of algal clumps varied from 20 to 150 g dry mass. Since it could influence coral health and subsequently fireworm abundance, it was included as a covariate where applicable. Differences among contact substrates (different coral species and dead coral) were assessed using PERMANOVA with substrate as fixed factor and algal dry mass as covariate. The effects of overgrown substrata (*M. faveolata* and dead coral) and sites on fireworm abundance were assessed using PERMANOVA with substrate as fixed factor, site as random factor and algal dry mass as covariate. All PERMANOVAs were based on Gower dissimilarity measures, appropriate for ecological data sets with many zero (Anderson et al. 2006). Differences in coral health scores were tested using an ordinal logistic regression including fireworm abundance in *H. opuntia* and algal dry mass as covariates. To determine whether the worms were aggregated, the frequency distribution of the number of worms per algal clump with 0 to four or more worms was tested for goodness of fit to a Poisson distribution with a *G* test after Williams's correction to avoid type I error (Sokal and Rohlf 1995). For this test, sites and coral species were pooled. However, separate analyses were run for algal clumps contacting live and dead corals ($n = 95$ and 51 clumps, respectively) to determine the influence of live coral on the worm behavior.

Data for the field experiment were analyzed using Kruskal-Wallis ANOVA on Ranks, followed by Student-Newman-Keuls post hoc tests on Ranks. Differences in the prevalence of white plague were analyzed using a heterogeneity *G* test, followed by a simultaneous test procedure (STP) to determine homogeneous groups (Sokal and Rohlf 1995). Data for the aquarium experiment were arcsine square root transformed to conform to assumptions of normality and homogeneity of variance and analyzed by a nested three-way ANOVA with plot (i.e., aquaria) as random factor, tank sections nested within plot and corallivores and algae treatments as fixed factors. Aggregative behavior was tested as above ($n = 12$ algal clumps, pooled aquaria).

Results

Field survey.—All the fireworms found in *H. opuntia* were identified as being *H. carunculata*, which also was the largest corallivore commonly present. The corallivorous snail *C. abbreviata* was found in only five of 146 collected *H. opuntia* clumps, compared to 71 clumps

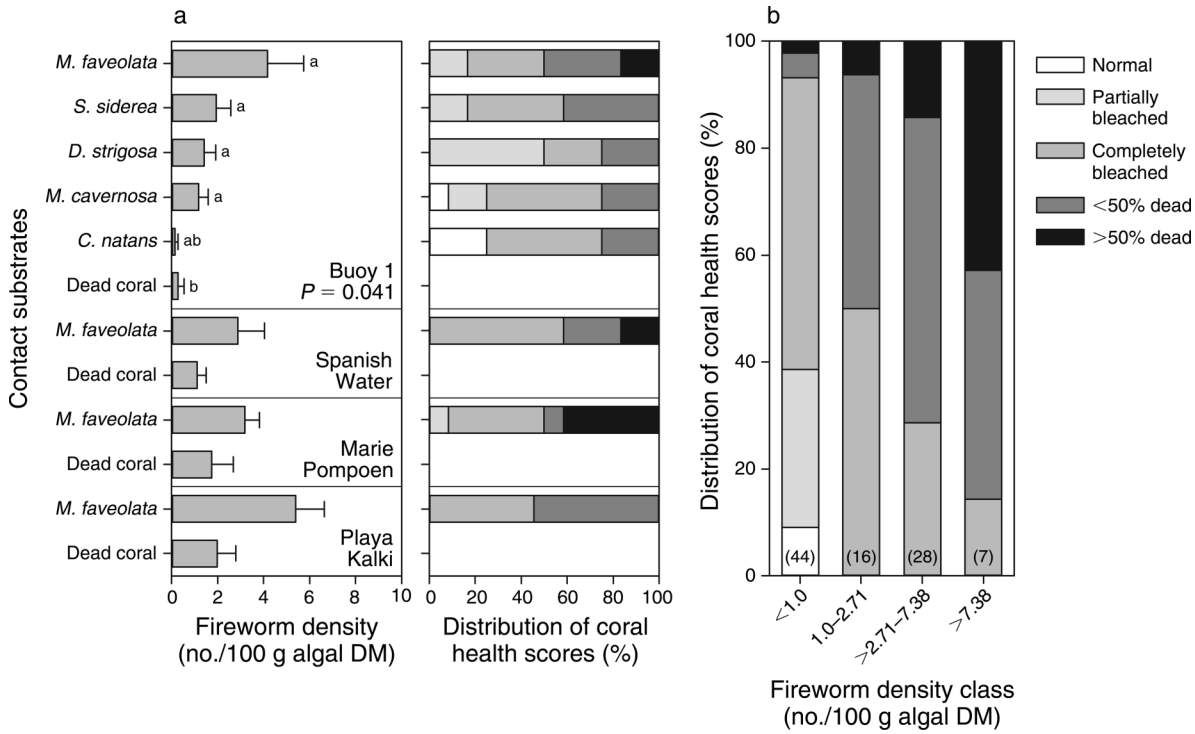

FIG. 1. (a) Fireworm abundance in *Halimeda opuntia* (means + SE, $n = 11$–14 replicates) and scores of coral health for all six contact substrates (*Montastraea faveolata*, *Montastraea cavernosa*, *Siderastrea siderea*, *Colpophyllia natans*, and *Diploria strigosa*) at Carmabi Buoy One and for *Montastraea faveolata* and dead coral skeleton at three other sites on Curaçao. Data on fireworm abundance at Carmabi Buoy One were analyzed by PERMANOVA. Letters indicate homogeneous subgroups by post hoc pairwise comparisons. (b) Scores of coral health observed underneath *H. opuntia* as a function of fireworm density (logarithmic scale). The number of interactions for each density class is in parenthesis. All coral species and sites were pooled. DM stands for dry mass.

containing *H. carunculata*. The negligible impact of *C. abbreviata* associated with *H. opuntia* was supported by the lack of relationship between the snail abundance in the clumps and tissue condition under the alga (ordinal logistic regression, $P = 0.264$) and the result of our laboratory experiment (see below). Thus, field data on *C. abbreviata* were not presented.

At Carmabi Buoy One, the abundance of *H. carunculata* in the algal clumps was highest in corals overgrown with large sized clumps and varied significantly among overgrown substrata (PERMANOVA, algal dry mass pseudo-$F = 18.21$, df = 1, $P = 0.001$; substrate pseudo-$F = 2.37$, df = 5, $P = 0.041$). There were more fireworms in *H. opuntia* overgrowing all coral species except *C. natans* than in algae overgrowing dead coral (Fig. 1a). The difference in fireworm abundances between *M. faveolata* and dead coral was consistent among sites, with site having no influence (substrate $P = 0.030$; site and substrate × site interaction $P > 0.05$; algal dry mass $P = 0.044$). Negative relationships between fireworm abundance and tissue condition under the algae were found for *M. faveolata* and *S. siderea* at Carmabi Buoy One (intercepts = 0.69 ± 0.31 and 0.78 ± 0.40 [mean ± SE], $P = 0.028$ and $P = 0.042$, respectively), with algal dry mass having no effect ($P > 0.05$). Pooling species and sites, both fireworm abundance and algal dry mass had a negative effect on coral health (intercepts = 0.46 ± 0.09 and 0.03 ± 0.01, $P < 0.001$, Fig. 1b). The frequency distribution of worms in the algal clumps overgrowing both live and dead corals significantly differed from a Poisson distribution (adjusted G test, $G_{adj} = 48.83$ and 27.48, respectively, df = 4, $P < 0.001$), demonstrating aggregative behavior in both cases. However, the G_{adj} value was markedly higher for live compared to dead coral, suggesting that this behavior is enhanced by the presence of live coral.

Field experiment.—The number of fireworms removed every five days in the *H. opuntia* added, epifauna removed (T2) treatment, averaged 1.09 ± 0.26 (mean ± SE) individuals per 100 g algal dry mass ($n = 100$ replicates) and did not differ among retrieval times (Kruskal-Wallis ANOVA on ranks, $H = 3.65$, df = 4, $P = 0.45$). At the end of the experiment, there were significantly more fireworms in the *H. opuntia* added (T1) treatment compared to all other treatments, including the epifauna removed (T2) treatment (Fig. 2a), indicating that the removal treatment effectively reduced the abundance of fireworms. Paralleling patterns of fireworm abundance, the percentage of recently dead tissue was significantly higher in the *H. opuntia* added (T1) treatment compared to all other treatments, including the epifauna removed (T2) and *H. opuntia*

mimic (T3) treatments (Fig. 2b and Appendix B: Fig. B1). These results suggest that the epifauna increased coral mortality in contact with living algae, but that this effect was not reproduced using mimic algae. The percentage of bleached tissue was highest in the epifauna removed (T2) treatment, followed by the mimic (T3) and *H. opuntia* added (T1) treatments (Fig. 2c). Thus, coral bleaching did not increase in response to the epifauna and seemed largely caused by physical mechanisms.

Signs of white plague were observed in six (30%) of the 20 transplants in the *H. opuntia*-added (T1) treatment, only one (5%) of the transplants in the epifauna-removed (T2) treatment, and in none of the transplants in the other treatments. Differences among treatments were significant (heterogeneity G test, $G_H = 39.58$, df = 4, $P < 0.005$, $n = 20$; Fig. 2d). STP tests indicated that {T1, T2} and {T2, T3, T4, T5} formed two homogenous groups. Thus, these results support a positive trend between the algae and coral disease, but we cannot conclusively attribute the disease to the sole effect of the algal-associated epifauna.

Laboratory experiment.—In the experiment with *H. carunculata*, coral mortality was affected by both *H. opuntia* and the presence of the predator, as indicated by the significant interaction between these factors (Fig. 3a, Appendix B: Fig. B2, Appendix C: Table C1). In the absence of algae, coral mortality was not recorded regardless of predator presence or absence. However, when corals were overgrown by algae, the presence of *H. carunculata* increased coral mortality (nested two-way ANOVA with plot and corallivores, $F_{1,20} = 4.97$, $P = 0.026$; Table C2). This effect did not occur with *C. abbreviata* (Fig. 3b). Coral bleaching was only affected by *H. opuntia*, with significantly increased bleaching with the algae present (Fig. 3c, d). Both corallivores had no influence on coral bleaching. Only one and two fragments in the experiments with *H. carunculata* and *C. abbreviata* respectively died, so no significant trends were detected in the occurrence of whole fragment mortality. The frequency distribution of *H. carunculata* in the algal clumps significantly differed from a Poisson distribution ($G_{adj} = 18.06$, df = 4, $P < 0.001$), demonstrating aggregative behavior. In contrast, *C. abbreviata* showed no aggregative behavior ($G_{adj} = 3.49$, df = 4, $P > 0.05$).

Discussion

This study suggests an interactive effect of predation by an invertebrate corallivore and overgrowth by an algal species on coral mortality. Contact with algae together with the addition of fireworms produced high coral tissue mortality. This mortality was significantly reduced in the presence of algae only, and no mortality occurred in the presence of fireworms only and when excluding both algae and fireworms. Both hypotheses regarding the distribution of *H. carunculata* in *H. opuntia* and their association with coral health in the field were validated, lending further support for the

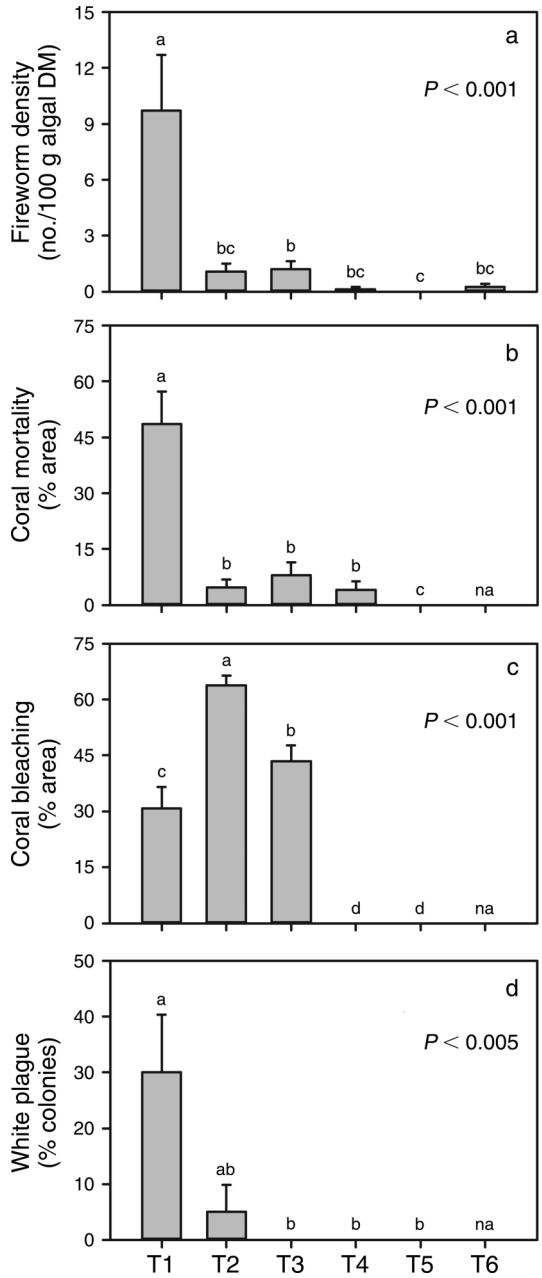

FIG. 2. Effect of algal-associated epifauna treatments on *Montastraea faveolata*. Treatments are: T1, *Halimeda* added, epifauna untouched; T2, *Halimeda* added, epifauna removed; T3, *Halimeda* mimic; T4, procedural control; T5, control; T6, dead coral. Treatments are more fully described in *Materials and methods: Field experiment*. (a) Fireworm abundance in mesh bags or experimental control area, (b) coral mortality, (c) coral bleaching, and (d) prevalence of white plague for the different treatments. Data were analyzed by Kruskal-Wallis ANOVA on ranks (a–c) or heterogeneity G test (d). Letters indicate homogeneous subgroups by post hoc Student-Newman-Keuls tests (a–c) or simultaneous test procedure (d). Values are means + SE ($n = 20$ colonies); na stands for not applicable.

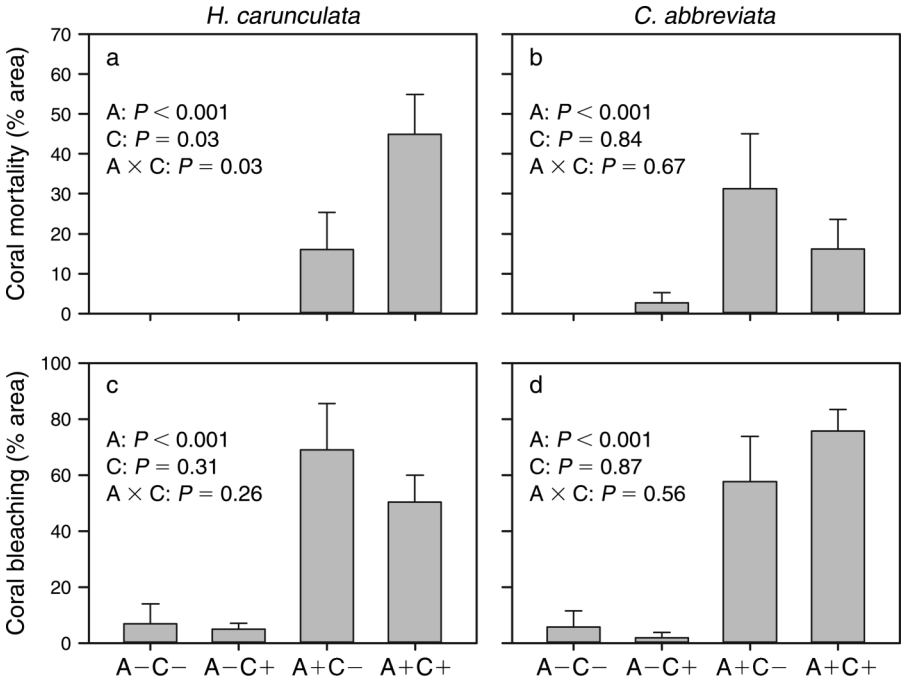

FIG. 3. Independent and combined effects of *Halimeda opuntia* and corallivores on (a, b) coral mortality and (c, d) bleaching. Data are for (a, c) *Hermodice carunculata* on *Montastraea faveolata* and (b, d) *Coralliophila abbreviata* on *Montastraea annularis*. Treatments are: C, corallivores treatment; A, algae (*H. opuntia*) treatment. Statistics are from a nested three-way ANOVA. Values are means + SE ($n = 4$ aquaria).

relevance of our experimental results. Fireworm abundance was higher in algae overgrowing the coral *M. faveolata* than in algae growing exclusively on dead coral, and increased fireworm abundance was associated with decreased coral health in the form of coral bleaching and mortality in the contact zone between corals and algae. Finally, the field experiment showed increased coral mortality in contact with *H. opuntia* in the presence of epifauna compared to the algae with epifauna removed. It is possible that epifauna other than *H. carunculata* contributed to this mortality. However, given the predominance of the worm in *H. opuntia*, its association with coral health and the results of our laboratory experiment, *H. carunculata* is likely to have been a major contributor.

In the field experiment, mimic algae did not produce the high coral mortality observed with living algae. There are two possible explanations. First, the algal mimic may not be the preferred habitat for *H. carunculata* and reduced worm densities in algae should, in principle, lower their effects on overgrown corals. While this explanation is plausible, fireworms, although not numerous, were found in the mimic algae and we observed them in a wide variety of other refuges, such as crevices, marine debris and other macroalgae. Alternatively, these results imply that algal-induced corallivory may be associated with chemical instead of physical effects produced by the algae. Although benthic algae can affect corals via a number of physical mechanisms (e.g., abrasion, shading [McCook et al. 2001]), coral mortality caused by algae, including *H. opuntia*, may require direct contact between both competitors and involve the release of algal exudates (Smith et al. 2006, Rasher and Hay 2010). Nearby (1–3 cm from colony edges) contact with *H. opuntia* produces no coral mortality (Vu et al. 2009). The most likely mechanism is that *H. carunculata* was initially attracted by dying coral tissues produced by algal-induced chemical effects during direct contact and thereafter both stressors acted in synergy to cause further coral mortality.

There is a precedent for corallivores to prefer damaged coral tissue as food source. For instance, the coral-feeding wrasse *Labrichthys unilineatus* has been shown to exhibit a strong preference for damaged coral tissue (McIlwain and Jones 1997). On Lizard Island, several species of corallivorous fishes selectively targeted disease lesions over adjacent healthy coral tissues (Chong-Seng et al. 2011). Such preferences may be caused by a release of olfactory attractants, a reduction in feeding deterrents (e.g., nematocysts) and/or higher energy rewards in damaged tissue (McIlwain and Jones 1997). *H. carunculata* possesses well-developed olfactory organs that could facilitate prey detection (Lizama and Blanket 1975). Since the algal mimics attracted low numbers of worms although inducing high coral bleaching, the presence of dying tissue as opposed to bleached tissue may be essential to this detection, and

metabolites leaking out of dying tissues could act as olfactory attractants.

Once corallivores are attracted to the zone of interaction, mechanisms leading to coral mortality remain to be elucidated. Gut contents of *H. carunculata* have been found to include coral tissue (Marsden 1963), confirming that the worm can feed on corals. Aggregative feeding behavior of predators, in particular scavengers, can cause high mortality in prey subject to even a small degree of stress and/or damage (Ramsay and Kaiser 1998) and constitutes an important mechanism driving trait-mediated indirect interactions (Werner and Peacor 2003). For example, Schmitt (1987) showed that adding bivalves to plots increased predation on gastropods due to aggregative behavior of invertebrate predators. In our study, the worms showed a marked aggregation behavior in *H. opuntia*, which appeared to be further enhanced by the presence of live coral. *Halimeda* alone is able to damage and weaken corals, as confirmed by our algal treatments without corallivores and laboratory work (Smith et al. 2006). The alga may thus lead to high coral mortality by enhancing the aggregative behavior of *H. carunculata*. The worms may also modify the physical, chemical and microbial microenvironment of overgrown corals, by excreting particulate wastes and leaving remains of decaying tissues that can then be colonized by microbes, increase oxygen demand, and reduce pH. These effects are likely to be exacerbated by the conditions of reduced flow underneath the algal canopy and to add on to the already detrimental effects of algae (Hauri et al. 2010). Moreover, a number of corallivores, including *H. carunculata*, have been implicated as potential disease vectors (Sussman et al. 2003, Williams and Miller 2005). Our field experiment supported a relationship between *H. opuntia* and white plague signs, reinforcing earlier work at the same site (Nugues et al. 2004). However, although trends were suggestive of a role of the algal-associated epifauna in the disease process, differences in disease prevalence between algal treatments with and without epifauna were not significant. Corallivores may also reduce coral disease progression through selective consumption of diseased coral tissues (Chong-Seng et al. 2011). Thus, their role in coral diseases is still equivocal.

The snail *C. abbreviata* was found in very low abundance in natural clumps of *H. opuntia* and our laboratory experiment provided no evidence of interaction between algal competition and corallivory by this species. *C. abbreviata* appears to be an obligate corallivore eating exclusively live coral (Ward 1965). In contrast, *H. carunculata* has been observed feeding on a variety of sessile organisms (Fauchald and Jumars 1979) and analyses of its gut contents indicated remains of corals, mollusks, polychaetes, and crinoids (Marsden 1963). Therefore, it appears as a facultative coral feeder that feeds on all kinds of food sources. Such a feeding mode may allow the worm to opportunistically switch feeding toward corals damaged by algae.

Our study demonstrates an indirect effect pathway by which algae can impact corals that shares similarities with apparent competition and habitat facilitation. In its classical definition, apparent competition occurs when an increase in the abundance of one species leads to a reduction in the abundance of a second species due to enhanced predation by a shared predator (Connell 1990; Fig. 1f in Werner and Peacor 2003). It is an important determinant of community structure in terrestrial (Chaneton and Bonsall 2000) and aquatic systems (Menge 1995). Although fireworms are unlikely to prey on *H. opuntia* and cannot be considered a shared predator, they contribute to an indirect effect that could be mistaken for a direct competitive effect. Likewise, facilitation by habitat-modifiers is common in ecological communities. In marine rocky intertidal habitats, Bertness et al. (1999) demonstrated that seaweeds facilitated green crabs, which in turn ate mussels under the algal canopy, closely resembling our study. Among the mechanisms involved, *Halimeda* might benefit *Hermodice* by increasing food supply and providing a refuge from predators.

Evaluating the impact of corallivory on coral reefs has been equivocal because many corallivores appear to cause sporadic damage to corals (Mumby 2009). We investigated one species of algae and acknowledge that the limited scope of our study prevents broad generalizations on the interactive effects between algal competition and corallivory. Nevertheless, numerous algal species harm corals (Rasher and Hay 2010) and thus could produce similar synergisms with corallivores. Interactive effects were not found for one (*C. abbreviata*) of the two experimentally studied corallivores. However, *H. carunculata* is distributed worldwide in tropical and warm temperate seas. Corallivores, in particular invertebrates that can easily find refuge in algae, form a highly diverse group (Rotjan and Lewis 2008). Invertebrates are often opportunistic feeders that due to their short life spans and large diet can cope with rapidly changing environments. To date, there is evidence that they can interact with acute disturbances such as hurricanes to compromise recovery of some coral species (Knowlton et al. 1990). Their interactions with chronic disturbances such as algal competition could be easily unnoticed due to their small size, cryptic habits, and little apparent damage and should be of concern in the face of accelerating and more severe disturbances on present-day reefs.

Acknowledgments

We thank the CARMABI foundation and staff, in particular M. J. A. Vermeij, P. Stokkermans, L. Pors, and C. Winterdaal for logistic support; K. Stolberg, J. Jachtner, D. Nugues and P. Kutter for field assistance; J. Claudet and W. Wosnick for statistical advice; and M. J. A. Vermeij and R. P. M. Bak for helpful discussions. This research was financially supported by the Leibniz Center for Tropical Marine Ecology.

Literature Cited

Anderson, M. J., K. E. Ellingsen, and B. H. McArdle. 2006. Multivariate dispersion as a measure of beta diversity. Ecology Letters 9:683–693.

Barott, K. L., and F. L. Rohwer. 2012. Unseen players shape benthic competition on coral reefs. Trends in Microbiology 20:621–628.

Baums, I. B., W. M. Miller, and A. M. Szmant. 2003. Ecology of a corallivorous gastropod. *Coralliophila abbreviata*, on two scleractinian hosts. I: Population structure of snails and corals. Marine Biology 142:1083–1091.

Bertness, M. D., G. H. Leonard, J. M. Levine, P. R. Schmidt, and A. O. Ingraham. 1999. Testing the relative contribution of positive and negative interactions in rocky intertidal communities. Ecology 80:2711–2726.

Chaneton, E. J., and M. B. Bonsall. 2000. Enemy-mediated apparent competition: empirical patterns and the evidence. Oikos 88:380–394.

Chong-Seng, K. M., A. J. Cole, M. S. Pratchett, and B. L. Willis. 2011. Selective feeding by coral reef fishes on coral lesions associated with brown band and black band disease. Coral Reefs 30:473–481.

Connell, J. H. 1990. Apparent versus "real" competition in plants. Pages 9–24 *in* J. B. Grace and D. Tilman, editors. Perspectives on plant competition. Academic Press, New York, New York, USA.

Fabricius, K. E., K. Okaji, and G. De'ath. 2010. Three lines of evidence to link outbreaks of the crown-of-thorns seastar *Acanthaster planci* to the release of larval food limitation. Coral Reefs 29:593–605.

Fauchald, K. 1977. The polychaete worms: definitions and keys to the orders, families and genera. Science Series 28:1–190.

Fauchald, K., and P. A. Jumars. 1979. The diet of worms: a study of polychaete feeding guilds. Oceanography and Marine Biology: An Annual Review 17:193–284.

Hauri, C., K. E. Fabricius, B. Schaffelke, and C. Humphrey. 2010. Chemical and physical environmental conditions underneath mat- and canopy-forming macroalgae, and their effects on understorey corals. PLoS ONE 5:e12685.

Knowlton, N., J. C. Lang, and B. Keller. 1990. Case study of natural population collapse: post-hurricane predation on Jamaica staghorn corals. Smithsonian Contributions to the Marine Sciences 31:1–25.

Lizama, J., and R. S. Blanket. 1975. Predation on sea anemones by the amphinomid polychaete, *Hermodice carunculata*. Bulletin of Marine Science 25:442–443.

Marsden, J. R. 1963. The digestive tract of *Hermodice carunculata* (Pallas) Polychaeta: Amphinomidae. Canadian Journal of Zoology 41:165–184.

McCook, L. J., J. Jompa, and G. Diaz-Pulido. 2001. Competition between corals and algae on coral reefs: a review of evidence and mechanisms. Coral Reefs 19:400–417.

McIlwain, J. L., and G. P. Jones. 1997. Prey selection by an obligate coral-feeding wrasse and its response to small-scale disturbance. Marine Ecology Progress Series 155:189–198.

McManus, J. W., and J. F. Polsenberg. 2004. Coral-algal phase shifts on coral reefs: ecological and environmental aspects. Progress in Oceanography 60:263–279.

Menge, B. A. 1995. Indirect effects in marine rocky intertidal webs: patterns and importance. Ecological Monographs 65: 21–74.

Miller, M. W., and D. E. Williams. 2006. Coral disease outbreak at Navassa, a remote Caribbean island. Coral Reefs 26:97–101.

Mumby, P. J. 2009. Herbivory versus corallivory: are parrotfish good or bad for Caribbean coral reefs? Coral Reefs 28:683–690.

Naim, O. 1988. Distributional patterns of mobile fauna associated with *Halimeda* on the Tiahura coral-reef complex (Moorea, French Polynesia). Coral Reefs 6:237–250.

Nugues, M. M., G. W. Smith, R. J. v. Hooidonk, M. I. Seabra, and R. P. M. Bak. 2004. Algal contact as a trigger for coral disease. Ecology Letters 7:919–923.

Pennings, S. C. 1997. Indirect interactions on coral reefs. Pages 249–272 *in* C. Birkeland, editor. Life and death of coral reefs. Springer, New York, New York, USA.

Ramsay, K., and M. J. Kaiser. 1998. Demersal fishing disturbance increases predation risk for whelks (*Buccinum undatum* L.). Journal of Sea Research 39:299–304.

Rasher, D. B., and M. E. Hay. 2010. Chemically rich seaweeds poison corals when not controlled by herbivores. Proceedings of the National Academy of Sciences USA 107:9683–9688.

Rotjan, R. D., and S. M. Lewis. 2008. Impact of coral predators on tropical reefs. Marine Ecology Progress Series 367:73–91.

Schmitt, R. J. 1987. Indirect interactions between prey: apparent competition, predator aggregation, and habitat segregation. Ecology 68:1887–1897.

Smith, J. E., et al. 2006. Indirect effects of algae on coral: algae-mediated, microbe induced coral mortality. Ecology Letters 9:835–845.

Sokal, R. R., and F. J. Rohlf. 1995. Biometry. Third edition. Freeman, New York, New York, USA.

Sussman, M., Y. Loya, M. Fine, and E. Rosenberg. 2003. The marine fireworm *Hermodice carunculata* is a winter reservoir and spring–summer vector for the coral-bleaching pathogen *Vibrio shiloi*. Environmental Microbiology 5:250–255.

Vu, I., G. Smelick, S. Harris, S. C. Lee, E. Weil, R. F. Whitehead, and J. F. Bruno. 2009. Macroalgae has no effect on the severity and dynamics of Caribbean Yellow Band Disease. PLoS ONE 4:e4514.

Ward, J. 1965. The digestive tract and its relation to feeding habits in the stenoglossan prosobranch *Coralliophila abbreviata* (Lamarck). Canadian Journal of Zoology 43:447–464.

Werner, E. E., and S. D. Peacor. 2003. A review of trait-mediated indirect interactions in ecological communities. Ecology 84:1083–1100.

Williams, D. E., and M. W. Miller. 2005. Coral disease outbreak: pattern, prevalence and transmission in *Acropora cervicornis*. Marine Ecology Progress Series 301:119–128.

Wootton, J. T. 2002. Indirect effects in complex ecosystems: recent progress and future challenges. Journal of Sea Research 48:157–172.

Supplemental Material

Appendix A

Schematic layout of the laboratory experiment (*Ecological Archives* E094-152-A1).

Appendix B

Photographs of treatments in field and laboratory experiments (*Ecological Archives* E094-152-A2).

Appendix C

Statistical output tables for the laboratory experiment (*Ecological Archives* E094-152-A3).